

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

**CM1 - Actuarial Mathematics for Modelling
(Paper A)**

May 2025

INDICATIVE SOLUTION

Q1 . c- The convexity of the asset cashflow series is equal to the convexity of the liability cashflow series (2)

$$\begin{aligned}
 \text{Q2. } EPV &= 10000 * (a_{20} + v^{20} * a_{60-20} p_{40}) \\
 &= 10000 * \left(\frac{1 - 1.04^{-20}}{0.04} \right) + (1.04)^{-20} * \frac{l_{60}}{l_{40}} * (\ddot{a}_{60} - 1) \\
 &= 10000 * (13.5903 + 0.4564 * 0.9423 * (14.134 - 1)) \\
 &= 192384
 \end{aligned}$$

Correct answer is d (2)

Q3.- a- i < iii < v < iv < ii (2)

$$\text{Q4. } A_{50} = 1 - d * \ddot{a}_{50} = 1 - (0.05/1.05) * \ddot{a}_{50}$$

$$\ddot{a}_{50} = 14.7$$

$$\bar{a}_{50} = \ddot{a}_{50} - 0.5 = 14.2$$

Correct answer is a. (2)

$$\text{Q5. Loan amount} = 450 * a_5 + 50 * Ia_5 \text{ at } 7\%$$

$$= 450 * 1845.09 + 50 * 11.75 = 2432$$

Correct answer is d. (2)

$$\text{Q6. Price} = D / I = 2 / 0.05 = 40$$

Correct answer is b. (2)

Q7. Differentiating wrt to i

$$P'(i) = - (D / (i^2))$$

$$P''(i) = 2D / (i^3)$$

$$\text{Convexity} = P''(i) / P = 2 / (i^2) = 800$$

Correct answer is c. (2)

Q8. Correct answer is c. Since investment return is same as discount rate no change in NPV (2)

$$\text{Q9. Accumulation} = 1000 * \exp \left(\int_0^{10} \delta(s) ds \right)$$

$$= 1000 * \exp \left[\int_0^8 0.07 - 0.005s ds + \int_8^{10} 0.06 ds \right]$$

$$= 1000 * \exp(0.40 + 0.12) = 1682$$

Correct answer is a. (2)

Q10. Correct answer is e. (2)

$$97.5 * (1+4.5\%)^{(t/360)} = 98.5$$

$$=((LN(98.5/97.5))/LN(1+4.5\%))*360$$

$$T= 83.46$$

Q11. $(1 + f_{3,2})^2 = \frac{(1+y_5)^5}{(1+y_3)^3} = \frac{(1+0.035)^5}{(1+0.03)^3} = 4.25\%$

Correct answer is b. (2)

Q12. Correct answer is a. (2)

$$EPV = 100000 * (v * q_{50} + v^2 * (p_{50} * q_{51}))$$

$$\text{From table book } q_{50} = 0.000545 \text{ and } q_{51} = 0.00063$$

Substituting these values in the equation, the solution comes out to be 107.

Q13. $4.75 p_{30.75} = 0.25 p_{30.75} * 4 p_{31} * 0.5 p_{35}$

$$= \frac{1 - q_{30}}{1 - 0.75 q_{30}} * (4 p_{31}) * (1 - 0.5 * q_{35})$$

$$= \left(\frac{1 - 0.00059}{1 - 0.75 * 0.00059} \right) * \left(\frac{9894.4299}{9919.3535} \right) * (1 - 0.5 * 0.000689)$$

$$= 0.9970$$

Correct answer is d (3)

Q14. $\sum_{1}^{\infty} k p_{20} = \sum_{1}^{\infty} \exp(- \int_0^k 0.015 dt) = \frac{\exp(-0.015)}{1 - \exp(-0.015)} = 66.17$

Correct answer is c (2)

Q15. $1 - (p_{[50]} * p_{[60]}) = 1 - \{(1 - 0.001971) * (1 - 0.005774)\} = 1 - 0.9923$

$$= 0.0077$$

Correct answer is a (2)

Q16. $a_{55|50} = (\ddot{a}_{50} - 1) - (\ddot{a}_{55:50} - 1) = (19.539 - 1) - (16.602 - 1) = 2.937$

Correct answer is b (2)

Q17. Standard Deviation $= \sqrt{75000^2 [{}^2A_{50} - (A_{50})^2]}$

$$= \sqrt{75000^2 * (0.06392 - 0.20508^2)} = 11089$$

Correct answer is e (2)

Q18. $P * \ddot{a}_{[40]:20} = 1000A_{[40]:20} + 0.54P + 0.06P\ddot{a}_{[40]:20}$

$$P = \left(\frac{1000 * 0.46423}{0.94 * 13.930 - 0.54} \right) = 36.98$$

Correct answer is d (2)

Q19. $V = 1000A_{45:15} - 0.94 * 36.98\ddot{a}_{45:15}$
 $= (1000 * 0.56206) - 0.94 * 36.98 * 11.386 = 166.3$

Correct answer is c (2)

Q20. Death benefit = $19 * 631 = 11989$

$$\text{Profit} = (17,095 + 631) * 1.045 - (11,989 * 0.015) - (0.985 * 18510) = 111.5$$

Correct answer is a (2)

Q21. Correct answer is c (2)

t	Profit	(t-1)px	Signature
1	170	1	170
2	-120	0.997492	-119.7
3	-7	0.99469	-6.96
4	15	0.991555	14.87

Q22. Correct answer is e (2)

Negative profit vector at time 3 will be zeroized at time2 = $-7v * p_{51} = -7 * (1 - 0.002809) / 1.04 = -6.71$

profit vector at time 2 after zeroizing = $-120 - 6.71 = -126.71$

Negative profit vector at time 2 will be zeroized at time1 = $-126.71v * p_{50} = -126.7118625 * (1 - 0.002508) / 1.04 = -121.53$

profit vector at time 1 after zeroizing = $170 - 121.53 = 48.47$

Profit signature at t=1 = $48.47 * 1 = 48.47$

Q23. Correct answer is c (2)

Property value after 22 years = $5000000 * (1 + 0.5\%)^{(22 * 2)} = 6,226,969$

$d^{(12)} = 12 * \{1 - (1 + i)^{(-1/12)}\} = 2.47\%$

$$\ddot{s}_{22}^{12} = \frac{(1 + i)^{22} - 1}{d^{(12)}} = 29.25$$

By principle of equivalence, $6,226,969 = 12X * 29.25$

Thus, $X = 17739.30$

Q24. Present value of an increasing annuity-due payable for 20 time units, the i th payment (of amount t) being made at time $t-1$.

Correct answer is c (2)

Q25. Correct answer is b (2)

Q26. Correct answer is a (2)

Q27. Correct answer is d (2)

Q28. Fund value after 25 years = $400 * \ddot{s}_{30}@i * 1.03^{20} + 400\ddot{s}_{20}@3\%$

Converting monthly rate in half yearly

$$1 + i = (1.005)^6$$

$i = 3.03775\%$ per half year

$$\ddot{s}_{30}@i = \frac{((1+i)^n - 1)}{d} = \frac{((1 + 3.03775\%)^{30} - 1)}{3.03775\%} * (1 + 3.03775\%) = 49.3215$$

$$\ddot{s}_{20}@3\% = \frac{((1 + 3\%)^{20} - 1)}{3\%} * (1 + 3\%) = 27.6765$$

Thus, fund value = $(400 * 49.3215 * 1.03^{20}) + (400 * 27.6765) = 35632.06 + 11070.60$

= INR 46,702.66

Correct answer is c (4)

Q29. $20 * (1+i)^3 + 5 * (1+i)^2 + 9 * (1+i) = 36.5$

When $i = 4\%$, value is 37.26

When $i = 3\%$ value is 36.43

Interpolating results in $i = 3.08\%$ i.e. 3.1% (rounded to 1 decimal)

Correct answer is b (3)

Q30. Present value of liabilities = $70500 v^6 + 300500 v^{15} @ 8\% = 139157$

$$DMT = \frac{70500 * 6 * v^6 + 300500 * 15 * v^{15} @ 8\%}{139157} = 12.13 \text{ years}$$

Correct answer is b (2)

Q31. Expected Revenue = $\left\{ \ddot{a}_3 * v * \frac{2}{3} \left(50 * \frac{3}{4} * + 25 * \frac{1}{4} \right) \right\} + \left\{ \ddot{a}_3 * v^2 * \frac{1}{3} \left(50 * \frac{3}{4} * + 25 * \frac{1}{4} \right) \right\}$

$$\ddot{a}_3 = 2.78$$

Expected Revenue = 109.96

$$\text{Expected cost} = 60 * 2/3 + (60+20v) * 1/3 = 66.17$$

$$\text{NPV} = 43.79 \text{ crores}$$

Correct answer is b (4)

$$\text{Q32. } P = 1.05^{\frac{1}{2}} * 0.75 * 4a_{10} + 110 v^{9.5} - 0.10(110 - P)v^{9.5}$$

$$P = 1.05^{\frac{1}{2}} * 3a_{10} + 99 v^{9.5} + 0.10Pv^{9.5}$$

Correct answer is a (3)

$$\text{Q33. } 2.5 * 120 / 107.3 = 2.80$$

Correct answer is d (2)

$$\text{Q34. } EPV = 40000 * A_{45:15}(1) + 50000 * v^{15} * \frac{l(60)}{l(45)} * A_{60}$$

$$\text{where, } A_{45:15} = 0.42556; A_{60} = 0.32692; l(60) = 9287.2164;$$

$$l(45) = 9801.3123$$

$$A_{45:15}(1) = A_{45:15} - v^{15} * \frac{l(60)}{l(45)} = 0.42556 - 0.417265 * \frac{9287.2164}{9801.3123} = 0.030181$$

$$EPV = (40000 * 0.030181) + (50000 * (1.06)^{-15} * 9287.2164 / 9801.3123 * 0.32692)$$

$$EPV = 7670.11$$

Correct answer is c (2)

$$\text{Q35. Premium} = 30,000 * (\ddot{a}_{65}(M) + \ddot{a}_{63}(F) - \ddot{a}_{65:63})$$

$$\text{From the tables, } \ddot{a}_{65}(M) = 13.666 \text{ and } \ddot{a}_{63}(F) = 15.606 \text{ and } \ddot{a}_{65:63} = 12.282$$

$$\text{Premium} = 30000 * (13.666 + 15.606 - 12.282) = 509700$$

Correct answer is d (3)

$$\text{Q36. } P \ddot{a}_{30} = 10,00,000 * \bar{A}_{30} + 10,000 * \bar{IA}_{30} + 500$$

$$\text{From table book, } \ddot{a}_{30} = 16.372; \bar{A}_{30} = 0.07328 * 1.06^{0.5}; \bar{IA}_{30} = 2.84692 * 1.06^{0.5}$$

$$16.372P = (1000000 * 0.07328 * (1.06^{0.5})) + (10000 * 2.84692 * (1.06^{0.5})) + 500$$

$$16.372P = 105257.213$$

$$P = 6429.10$$

Correct answer is c (3)

$$\text{Q37. Required provision is } 10,40,000 * \bar{A}_{33} + 15,000 * \bar{IA}_{33} - 6429.10 \ddot{a}_{33}$$

$$\text{From table book, } \ddot{a}_{33} = 21.354; \bar{A}_{33} = 0.17868 * 1.04^{0.5}; \bar{IA}_{33} = 7.25340 * 1.04^{0.5}$$

$$\text{Provision} = (1040000 * 0.17868 * (1.04^{0.5})) + (15000 * 7.2534 * (1.04^{0.5})) - (6429.10 * 21.354)$$

$$= 163,176$$

Correct answer is b

(3)

Q38. $DSAR = 60000 * 1.04^{0.5} - 11354.90 = 49,833.33$

$$EDS = DSAR * 1900 * q_{70} = 49,833.33 * 1900 * 0.024783 = 23,46,537$$

$$ADS = 33 * DSAR = 16,44,500$$

$$\text{Mortality profit} = EDS - ADS = 702,037$$

Correct answer is a

(4)

Q39.

Particulars	Year 1	Year 2	Year 3	Year 4
start of the year	-	95,187	203,095	316,905
Allocated premium	95,000	102,500	102,500	102,500
Bid offer spread	(4,750)	(5,125)	(5,125)	(5,125)
Interest	5,415	11,554	18,028	24,857
Management fee	(478)	(1,021)	(1,592)	(2,196)
Value at the end	95,187	203,095	316,905	436,942

Correct answer is a

(4)

Q40.

Particulars	Year 1	Year 2	Year 3	Year 4
UnAllocated premium	5,000	(2,500)	(2,500)	(2,500)
Bid offer spread	4,750	5,125	5,125	5,125
Expenses	2,500	500	500	500
Interest	312	91	91	91
Management fee	478	1,021	1,592	2,196
Extra death benefit	593	482	330	127
End of year cashflow	7,447	2,755	3,479	4,285

Correct answer is d (4)

Q41. Correct answer is b (4)

$$P\ddot{a}_{55:10} = 49000 * A_{55:10}(1) + 1000 (IA)_{55:10}(1) + 60000 * \frac{l(65)}{l(55)} * v^{10}$$

$$\text{Where, } l(65) = 8821.2612 \quad l(55) = 9557.8179$$

$$A_{55} = 0.26092; A_{65} = 0.40177; (IA)_{55} = 5.22868; (IA)_{65} = 5.50985;$$

$$\ddot{a}_{55:10} = 7.610$$

$$A_{55:10}(1) = A_{55} - \frac{l(65) * v^{10}}{l(55)} * A_{65}$$

$$= 0.26092 - ((1.06)^{-10} * 0.40177 * 8821.2612 / 9557.8179) = 0.05386$$

$$(IA)_{55:10}(1) = (IA)_{55} - \frac{l(65) * v^{10}}{l(55)} * ((IA)_{65} + 10A_{65})$$

$$= 5.22868 - ((1.06)^{-10} * 8821.2612 / 9557.8179) * (0.40177 * 10 + 5.50985) = 0.3185$$

Thus,

$$7.61P = (49000 * 0.05386) + (1000 * 0.3185) + (60000 * ((1.06)^{-10}) * 8821.2612 / 9557.8179)$$

$$7.61P = 33879.58$$

$$P = 4451.98$$
